VIP Individual Project Part 2: Beam Design and AI Model Comparison

Experiment status

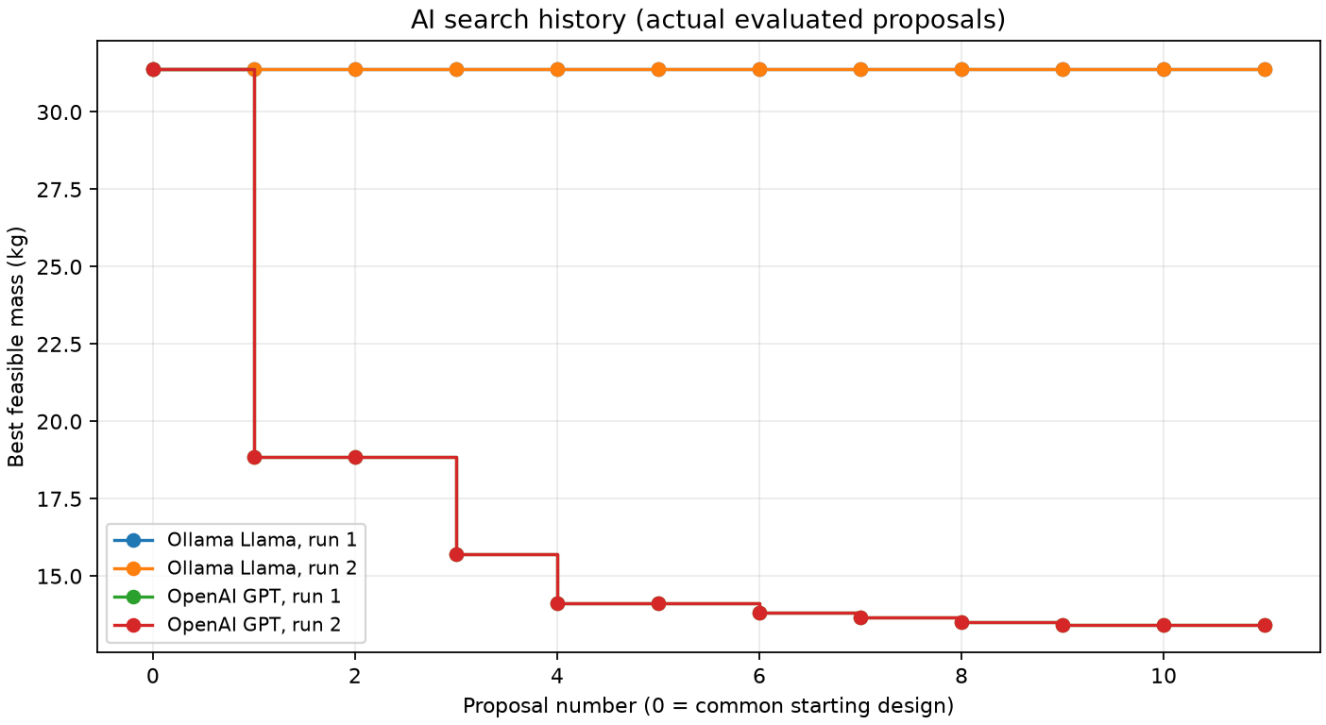
Complete experimental records

All mass, stress and deflection values come from the instructor-supplied beam_model.py. Dimensions are passed in millimeters, following the callable and assignment example.

Results comparison

Method / model / run	Best kg	Width mm	Height mm	Defl. margin mm	Stress margin MPa	Evals	Invalid	Time s
SciPy SLSQP / scipy.optimize.minimize / 1	13.4233	20.000	85.499	0.0000	17.401	152	10	0.01
Ollama Llama / llama3.2:1b / 1	31.4000	50.000	80.000	0.5117	73.125	12	0	45.07
Ollama Llama / llama3.2:1b / 2	31.4000	50.000	80.000	0.5117	73.125	12	0	29.32
OpenAI GPT / gpt-4.1-mini / 1	13.4235	20.000	85.500	0.0000	17.404	12	0	10.64
OpenAI GPT / gpt-4.1-mini / 2	13.4235	20.000	85.500	0.0000	17.404	12	0	9.09

AI search histories



Numerical optimizer

SLSQP convergence success: True. Message: Optimization terminated successfully. Actual evaluator calls, counting objective, constraint and final calls: 152.

Comparison and limitations

The shared evaluator uses ideal small-deflection beam theory and excludes contact effects, self-weight and impacts. SciPy selects candidates numerically; each AI model sees prior evaluations and proposes the next design. Invalid model replies consume proposal opportunities and receive no beam evaluation. Model quality cannot be concluded without all four actual AI runs. Run-to-run variation, repeated designs and improvement from feedback must be interpreted from completed logs. This comparison is not an Ansys finite-element comparison.

Reproducibility

Common starting design: 50 x 80 mm. Each agent run: 11 proposals, temperature 0.2, shared prompt and strict JSON validator. Requested Ollama identifier: llama3.2:1b; requested OpenAI identifier: gpt-4.1-mini. Python packages: scipy, matplotlib, reportlab; Ollama local server and OpenAI API key required. GitHub repository: [ADD YOUR ACTUAL REPOSITORY URL BEFORE SUBMITTING].